Description | Accepted | Editorial | Solutions | Submissions

878. Nth Magical Number

Solved

Hard | Topics | Companies

A positive integer is *magical* if it is divisible by either `a` or `b`.

Given the three integers `n`, `a`, and `b`, return the nth magical number. Since the answer may be very large, **return it modulo** $10^9 + 7$.

Example 1:

```
Input: n = 1, a = 2, b = 3
Output: 2
```

Example 2:

```
Input: n = 4, a = 2, b = 3
Output: 6
```

Constraints:

- $1 <= n <= 10^9$
- $2 <= a, b <= 4 * 10^4$

Discover more | Math | Mathematics

1.3K | 16 | 3 Online

Code

C++ | Auto

```cpp
class Solution{
public:
    long long gcd(long long a,long long b){
        while(b){
            long long t=b;
            b=a%b;
            a=t;
        }
        return a;
    }
```

Saved | Ln 9, Col 18

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 | Case 2

Input

n =
1

a =
2

b =
3

Problem List | Submit | 0 | Premium

Description | Accepted | Editorial | Solutions | Submissions

All Submissions

Accepted 70 / 70 testcases passed

Raghav_Yadav submitted at Feb 17, 2026 15:14

Editorial | Solution

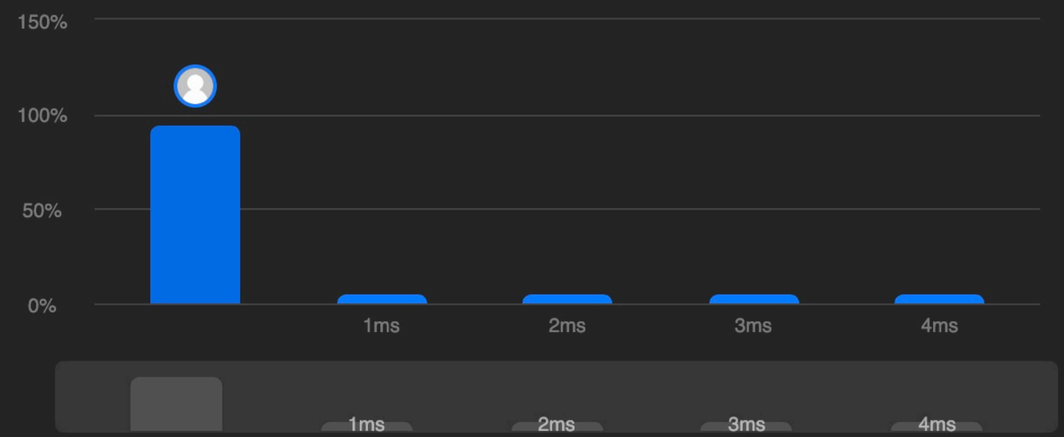
Runtime

0 ms | Beats 100.00%

Analyze Complexity

Memory

7.79 MB | Beats 80.62%



Code | C++

```cpp
class Solution{
public:
    long long gcd(long long a,long long b){
```

Code

C++ | Auto

```cpp
class Solution{
public:
    long long gcd(long long a,long long b){
        while(b){
            long long t=b;
            b=a%b;
            a=t;
        }
        return a;
    }

    int nthMagicalNumber(int n,int a,int b){
        const long long MOD=1000000007;
        long long l=min(a,b);
        long long r=(long long)n*min(a,b);
        long long lcm=(long long)a*b/gcd(a,b);
        while(l<r){
            long long mid=l+(r-l)/2;
            long long cnt=mid/a+mid/b-mid/lcm;
            if(cnt<n) l=mid+1;
            else r=mid;
        }
        return l%MOD;
```

Saved | Ln 9, Col 18

Testcase | Test Result